

BNI Enterprises – Bike Dealer Management System

Comprehensive Feature & Security Report

System Overview

A complete PHP-based Single-Page Application for bike dealerships. Manages purchase orders, inventory, sales (cash & installment), returns, customer/supplier ledgers, accessories, quotations, payments, income/expenses, and advanced reporting. Features role-based access, CAPTCHA, IP-banning, CSRF, session timeouts, and a responsive themeable UI.

Module-Wise Feature Breakdown

 Module & Features	 Key Capabilities	 Security & Access
 Dashboard & Analytics ✓ Live summary cards (In Stock, Sold, Returned, Damaged/Lost) ✓ Financial cards (Purchase Value, Sales Value, Tax, Profit, Expenses, Pending Cheques, Today's Sales) ✓ Overdue installment & pending cheque alerts ✓ Interactive charts: Sales Trend (line), Model-wise Stock (doughnut), Income vs Expense (bar), Inventory Status (pie) built with Chart.js ✓ Quick-action buttons (New Sale, New Purchase, Add Customer, etc.) ✓ Model-wise stock summary table ✓ Recent 10 sales & purchases ✓ Sidebar collapsible, dark/light theme toggle	 Real-time financial overview  Visual analytics for decision-making  One-click navigation to core operations  Fully responsive layout	 Dashboard access controlled by role (can_view)  Data visibility limited to user's permissions  CSRF token included in quick-action forms
 AI-Enabled Analytics ✓ Interactive charts powered by Chart.js (no external AI, purely client-side)	 Data-driven insight without third-party services	 Charts only render for allowed pages
 Purchase Entry ✓ Single form to create purchase order with multiple bikes ✓ Dynamic bike rows (chassis, motor, model, color, purchase price, safeguard notes, image) ✓ Dynamic payment rows (cash, cheque, bank transfer, online) with cheque fields ✓ Auto-calculate purchase total vs payment total, auto-divide payment among bikes ✓ Supplier selection with inline "Add Supplier" modal ✓ Model selection with inline "Add Model" modal ✓ AJAX chassis uniqueness check	 Rapid entry of bulk bike purchases  Photo documentation per unit  Flexible payment splitting  Live summary of balances	 Only users with purchase add/edit permission ✓ Required fields validated on client & server  Supplier & model integrity checks  Image type/size validation

- ✓ Image upload per bike (resized to max 800px, JPEG 75%)

Inventory / Stock

- ✓ Full table listing all bikes (with DataTables responsive, column filtering, state saving)
- ✓ Powerful filter bar: search by chassis/motor/model/color, status (all statuses including damaged/lost), model, color, date range
- ✓ Bulk actions: delete selected, export selected to CSV, print
- ✓ Delete bike with permission (prevents deleting sold/returned/damaged bikes)
- ✓ Edit bike modal (color, purchase price, status, notes, image)
- ✓ View bike detail page with timeline (purchased → inventory → sold → returned) and full details
- ✓ Bike image display
- ✓ Page totals for purchase price, selling price, margin

-  Advanced inventory filtering
-  Complete bike lifecycle visibility
-  Photo-documented units
-  Export/print capabilities

-  Role-based access for view, edit, delete
- ✓ Edit actions log status changes (e.g., marking damaged triggers expense entry)
- ⊘ Delete restricted to non-terminal statuses
-  CSRF on all POST actions

Sales Entry

- ✓ Sell single bike from inventory (chassis search with details auto-fill)
- ✓ Selling price input, auto-calculated tax and margin (supports tax on purchase or selling price)
- ✓ Customer selection with inline “Add Customer” modal
- ✓ Customer filer status display (affects tax? – shown but tax logic is simple)
- ✓ Down-payment entry with payment method (cash, cheque, bank transfer, online); cheque fields appear dynamically
- ✓ Accessories sold with chosen bike: dynamic rows (accessory, qty, unit price, discount, final price) – stock deduction
- ✓ Full installment plan creation: total installments, installment amount, first due date; auto-calculate from remaining balance
- ✓ Walk-in customer restriction: must pay full amount upfront (no partial payment)
- ✓ Upon sale: bike status → ‘sold’, margin recorded, payment entry, ledger entries (debit for sale, credit for down-payment), installment records created
- ✓ Print invoice (invoice number, company/branch, customer details, bike details, accessory breakdown, payment summary)

-  Complete point-of-sale workflow
-  Supports cash, cheque, installment sales
-  Stock auto-deduction for accessories
-  Professional invoice generation
-  Margin & tax calculated instantly

-  Only users with **sale** add permission
- ✓ Required fields validated
-  Ledger automatically created
- ⊘ Walk-in customer cannot leave balance
-  Invoice respects **show_purchase_price** setting

↔ Returns

- ✓ Two sub-tabs: **Sales Returns** (return a sold bike to customer) and **Purchase Returns** (return an in-stock bike to supplier)
- ✓ **Sales Return**: select sold bike, return date, refund amount, refund method (cash/cheque/bank/online with cheque fields), notes
- ✓ Updates bike status to **returned**, records payment (customer_refund), creates reversal ledger entries
- ✓ **Purchase Return**: select in-stock bike, return date, refund received amount, method; updates status to **returned_to_supplier**, records payment (supplier_refund), creates reversal ledger entries
- ✓ Auto-fills refund amount from purchase price for purchase return

-  Handles both customer & supplier returns
-  Full accounting reversals
-  Audit trail via ledger

-  Only users with **returns** add/edit permission
- ✓ Status validation (only sold bikes for sales return; in-stock for purchase return)
-  All transactions recorded in ledger

💰 Payments Register

- ✓ Lists all payments with filters: cheque status (pending/cleared/bounced/cancelled), transaction type, bank name, date range
- ✓ Summary stats: pending, cleared, bounced, cancelled
- ✓ Per-payment actions: mark cheque as **cleared** or **bounced** (bouncing reverses amount from installment / customer ledger and adds penalty via ledger), delete payment
- ✓ Bounced cheque handling: involved ledger adjustments and installment reversal
- ✓ All payment types (purchase, sale, installment, expense, supplier payment, customer refund)

-  Complete cheque management
-  Workflow for cheque clearance
-  Real-time financial status by method

-  Only users with **payments** edit/delete permissions
- ✓ Status change triggers financial corrections
-  All actions logged in ledger

Installments

- ✓ List of all installments with filters: status, customer, due date range
- ✓ Summary columns: due date, customer, chassis, model, installment amount, amount paid, penalty, status
- ✓ "Pay" button on pending/overdue installments opens modal: enter payment amount, penalty fee, payment date, payment method (cash/cheque/bank/online with cheque fields)
- ✓ Upon payment: updates installment (paid amount, penalty, status), records payment (transaction_type = installment), creates ledger entries (credit for payment, debit for penalty)

-  Flexible installment payment recording
-  Due-date tracking
-  Penalty fee support

-  Only users with **installments** edit permission
- ✓ Validates amount paid ≤ remaining due
-  Updates ledger automatically

Quotations

- ✓ Create/edit quotation: date, validity, customer

-  Pre-sales quoting
-  One-click

-  Only users with **quotations** add/edit/delete

 Module & Features	 Key Capabilities	 Security & Access
(with details display), bike, quoted price, optional accessories (same dynamic accessory rows), notes ✔ Convert a pending quotation to a sale: automatically marks bike as sold (creates all ledger/payment/installment entries as per sale workflow) and sets quotation status to converted ✔ List all quotations with status (pending/accepted/rejected/converted), edit, delete	conversion to actual sale 🔧 Accessories included in quote	permission 🔄 Conversion requires sale add permission ✔ Status-based restrictions
 Income / Expense Tracker ✔ Separate module for non-bike income/expenses ✔ Filter by date range, type, category ✔ Summary cards: total income, total expense, net balance, avg transaction, top income/expense categories ✔ Charts: income by category (doughnut), expense by category (doughnut), daily trend (line) ✔ Add/edit/delete entries with date, type, category (auto-suggest datalist from existing), amount, payment method, reference, notes	📊 Independent financial tracking 📈 Visual category breakdown 📄 Categorization for expense analysis	🔒 Only users with income_expense add/edit/delete permission ✔ Validation of required fields
 Customer Ledger ✔ Select customer from dropdown, view complete ledger: all debit/credit entries with running balance ✔ Summary cards: total billed, total paid, remaining balance (due/advance) ✔ Actions: receive payment (modal), make payment (advance/loan) – both create payment records and ledger entries ✔ Purchase history: table of bikes bought by customer ✔ Printable ledger	📁 Full customer accounting 🏦 Advance/loan tracking 🖨️ Print-ready ledger	🔒 Viewable only with customer_ledger view permission
 Reports ✔ Sub-tabs with dedicated filters (from-to, year, month): • Current Stock: lists all in-stock bikes with days in stock, total value • Sold Bikes: sold bikes within date range with profit/tax • Model-wise Sales: aggregated per model (inventory, sold, available, returned, damaged/lost, totals) • Tax Report: monthly tax summary • Profit/Margin: monthly profit with avg margin • Bank/Cheque Report: cheque payments by bank and transaction type • Monthly Summary: units purchased vs sold per month	📊 Comprehensive business intelligence 📅 Date-driven drill-downs 🖨️ Printable reports 📄 Export-ready (implicit via print)	🔒 All reports require reports view permission ✔ Filters sanitized to prevent SQL injection

- **Daily Ledger:** for a selected date shows sales, inventory added, expenses, other income
- **Purchase vs Sales:** month-wise comparison
- **Accessory Stock:** current stock, purchase/selling value
- **Installments Summary:** per-customer installment totals, overdue balances
- **Money by Destination:** per-destination allocation totals with date filtering
- **Money by Sale:** per-sale breakdown showing destination allocation badges
- **Untracked Sales:** sales with no/partial money allocation, with direct Track action
- **Money Flow:** monthly money flow per destination type (bank/person/wallet)
- ✓ Date/year/month filters per report
- ✓ Print-optimized layouts for all reports

Money Destinations

- ✓ CRUD for money destinations (bank, person, wallet)
- ✓ Name, type, details, active/inactive toggle
- ✓ Summary cards showing count per type
- ✓ Total allocated amount per destination
- ✓ Delete protection if allocations exist

- 💰 Centralized money destination management
- 🔒 Referential integrity enforcement
- 📱 9 default destinations pre-seeded

- 🔒 CRUD requires **money_destinations** permissions
- ✓ Cannot delete destinations with linked allocations

Money Tracking

- ✓ Track where sale money goes (bank/person/wallet) per sold bike
- ✓ Multiple allocations per sale supported
- ✓ Filter by sale or destination
- ✓ Remaining untracked amount shown per sale
- ✓ Optional inline tracking during sale entry
- ✓ Full CRUD on allocations at any time after sale
- ✓ Audit trail: tracks who created each allocation

- 📊 Sale-level money allocation management
- 📱 Untracked balance visibility
- 🔍 Filterable allocation listing
- ♻️ Supports post-sale allocation CRUD

- 🔒 Requires **money_tracking** permissions
- ✓ ON DELETE CASCADE cleans up on sale deletion
- 📄 Created-by audit trail per allocation

Models Management

- ✓ CRUD for bike models (code, name, category, short code, image)
- ✓ List with count of total inventory, in stock, sold
- ✓ Quick actions: purchase (redirect to purchase entry with model pre-selected), sell
- ✓ Image upload per model (resize to 800px)

- 🏷️ Model catalog with visual reference
- 📊 Model-wise inventory stats

- 🔒 Only users with **models** add/edit/delete permission
- ✓ Deletion blocked if bikes linked to model

Accessories Management

- ✓ CRUD for accessories (name, SKU, purchase price, selling price, current stock)
- ✓ Advanced reporting cards: total items, total stock, stock value (PP/SP), total sold qty, revenue, profit,

- 💎 Full accessory inventory
- 📊 Profitability analysis

- 🔒 Only users with **accessories** add/edit/delete permission
- ✓ Delete blocked if sold in sales

Module & Features

discounts

✓ Charts: Top 5 Sold Accessories (bar), Stock & Sales overview

✓ Stock deduction on sale; validation on delete if linked to sales

Key Capabilities

 Real-time stock control

Security & Access

Customers & Suppliers

✓ Standard CRUD with search

✓ Show bike purchase count and total amount per customer; order count and total purchase value per supplier

✓ Quick-link to customer/supplier ledger

✓ Delete blocked if linked to bikes/purchase orders

 Entity management with usage stats

 Direct ledger access

 Needs respective add/edit/delete permissions

Users & Roles

✓ User management: username, full name, role, password (strong password enforcement), active/inactive

✓ Role management: define roles with description

✓ Granular permissions matrix: for each page (dashboard, inventory, purchase, sale, etc.) set view, add, edit, delete individually

✓ Administrator role cannot be deleted or renamed

✓ Users cannot delete themselves or admin

 Multi-user with fine-grained control

 Role-based UI navigation (nav shows only allowed pages)

 All actions require **users/roles** add/edit/delete permissions

 Password complexity: min 8 chars, one uppercase, one lowercase, one digit, one special character

✓ Admin role always has full access

Settings

✓ Company name, branch, currency, tax rate (0.0-1.0), tax calculation base (purchase or selling price), show purchase price on invoice toggle

✓ Session timeout settings: idle (seconds) and absolute (seconds)

✓ Change admin password (with strength validation)

✓ Database backup: full SQL dump download

✓ Database restore: upload .sql file with warning

✓ System info: version, author, PHP/MySQL versions, database name, server time

 Central configuration

 One-click backup/restore

 Configurable session security

 Only users with **settings** edit/view permission

 Password change uses current session user

 Restore requires confirmation and validates SQL

Authentication & Security

✓ Login with math CAPTCHA (SVG image, lifetime 5 min)

✓ CSRF token on every POST form (validated, regenerated on session)

✓ IP-based temporary ban after 7 failed login attempts (3-hour ban, stored in JSON in temp dir)

✓ Session cookies: HttpOnly, Secure possible (but not forced), SameSite=Strict, use_only_cookies

✓ Session timeout: idle (default 40 min) and absolute (default 8 hours) configurable

✓ Password hashing: bcrypt

✓ Role-based access control: every page and action

 Enterprise-grade security layers

 Mathematical CAPTCHA to prevent bots

 IP banning after repeated failures

 Automatic session expiry

 All security measures are server-enforced

 Failed login attempts logged to JSON

 Password complexity enforced for new users

can be restricted

- ✓ Secure image upload: file type, size (max 400KB), resize+compress to JPEG
- ✓ AJAX endpoints protected by CSRF token inclusion

UI/UX & Responsive Design

- ✓ Dark/Light theme with CSS variables, toggle button
- ✓ Collapsible sidebar (persistent via localStorage on desktop, overlay on mobile)
- ✓ Smooth animations (animate.css)
- ✓ SweetAlert2 for confirmations, alerts, and form validation feedback
- ✓ JustValidate for client-side form validation
- ✓ Select2 for enhanced dropdowns with search (except DataTables length menus)
- ✓ DataTables with responsive, paging, search, state save
- ✓ Print-optimized stylesheet (hides sidebar, topbar, etc.)
- ✓ Accessible keyboard-friendly inputs
- ✓ Toast notifications for success/error messages

 Mobile-first, fully responsive

 Accessible and user-friendly

 Instant feedback on actions

 Rich interactive tables

 CSRF tokens added to all dynamic forms via jQuery

 Client-side validation before server submission

Export & Print

- ✓ Inventory CSV export (bulk selection or full filtered view)
- ✓ Database full backup as SQL download
- ✓ Print invoice and various tables with print-specific CSS

 Data portability

 Professional hard-copy output

 Only accessible to users with appropriate view permissions

Auto-Install & Seed Data

- ✓ Single-click database installation: creates all tables, default settings, admin user, roles, permissions, sample models (13 electric bikes), accessories, customers, suppliers
- ✓ Auto-detects missing DB and shows setup screen

 Rapid deployment without separate installer

 Install only possible if DB not yet created

Multi-Language / Localization

- ✓ Not explicitly multi-language, but currency & date formats configurable in settings
- ✓ All date displays use `fmt_date()` (d/m/Y)
- ✓ Money formatted as per currency symbol

 Regional adaptability

System Highlights

Category

Key Achievements

Security

bcrypt hashing, CSRF tokens, IP-based brute-force ban, math CAPTCHA, session hardening, role-based permissions with granular CRUD, SQL injection prevention via prepared

 Category	 Key Achievements
	statements
 Completeness	Full cycle: purchase → inventory → sale (cash/installment) → return → ledger. Accessories, quotations, expenses, multi-supplier/customer management, installment tracking
 Business Intelligence	Real-time dashboard cards, 4 interactive charts, 15 report sub-modules with drill-downs, model-wise and customer/supplier-wise analytics, money destination tracking and flow reports
 Financial Accuracy	Double-entry ledger for sales, payments, returns, cheque bounce. Automatic margin/tax calc. Installment penalty handling. Income/expense separate tracking.
 User Experience	Dark/light theme, collapsible sidebar, data tables with inline search/pagination, SweetAlert dialogs, client-side validation, print-ready invoices
 Performance	Single-file architecture, minimal dependencies, optimized image upload resizing, DataTables state saving, efficient SQL queries
 Customizability	Configurable tax rate (on purchase/selling), invoice display options, session timeouts, company details, roles and permissions per page and action

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Empowering dealerships with smart, secure, and user-friendly business management